CSA1737 - AI Col - AT3

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1) Types of Intelligent Agents :-

(a) simple Reflex Agent

- works on condition - action rules.
- Uses only current percept (no memory)
- Eg :- If motion detected → turn on light.
- In home automation : can handle basic tasks like turning on/off lights when motion is detected.

(b) model - Based Agent

- maintains an internal state of the environment.
- Uses a model to predict future states.
- Example : If temperature is rising, it predicts higher cooling need.
- In home automation :- Learns room conditions and adjusts AC/fan even without immediate sensor input.

(c) Goal - Based Agent :-

- Acts to acheive specific goals.
- Uses search and planning to find best actions.
- Example :- maintain temperature within 22-25°C
- In home automation : plans actions to reach comfort level based on time of day and user schedule.

(d) Utility - Based Agent

- Chooses the best action based on a utility function.

- Considers multiple factors (comfort, energy, security).

Eg:- Adjusts brightness and AC to maximize comfort while minimizing energy cost.

2. How each type works in the system

Agent Type	Function in smart Home.
Simple Reflex	Quick response to immediate events (e.g:- motion → light ON)
Model - Based	uses past data and environment model (Eg:- remembers room temperature, user habits)
Goal - Based	Plans actions to met goals (Eg:- maintain desired temp, ensure security at night)
Utility - Based	selects best action considering comfort, energy & security.

3. Which approach is most effective?

A hybrid approach combining all four types is most effective. A utility - based agent with a model and goal - based planning ensures : comfort, energy, and security.

Conclusion:-

A hybrid intelligent agent (utility + goal + model, with simple reflex) is the most effective for ensuring comfort energy, efficiency and security in a smart home automation system.

2) (i) problem description

- the robot is in an unknown maze with no prior knowledge of the layout.
- this is a search problem where the goal is to reach the exist from the start state.

2. Uniformed search strategies

(a) uniform cost search (UCS)

- Expands the node with the lowest path cost first.
- works well when each step has a different cost.
- guarantees the optimal solution.
- uses a priority queue to select next node.

(b) Iterative Deepening search (IDS)

- Combines DFS and BFS.
- Repeatedly increases the depth limit (0,1,2 ---).
- Finds the shallowest solution.
- uses less memory than BFS.

3) Comparision :- time and space complexity :-

Algorithm	time complexity	space complexity	optimal ?
UCS	$O(b^{1+(C^*/\varepsilon)})$	$O(b^{1+(C^*/\varepsilon)})$	yes
IDS	$O(b^d)$	$O(b^d)$	yes (if step cost = 1)

4. Agent Types in this problem

- Simple reflex Agent:-
  - Uses current sensor input (e.g:- wall, open path)
  - No memory → may get stuck in loops.
- model - Based Agent
  - maintains a map of explored areas.
  - Avoids revisiting the same path.
- Goal - Based Agent
  - plans path to reach exit using search strategy.
  - Can backtrack if needed.
- Utility - Based Agent
  - considers path cost (time, energy, risk)
  - chooses safest and shortest path.

5) Best Combination & Justification

model- based agent + UCS (or IDS if memory is limited.

Reason:- model - based agent remembers explored areas and avoids loops. UCS ensures the lowest cost path. this combination ensures both efficient decision - making and optimal/near-optimal Path.

Conclusion:-

UCS and IDS help the robot explore the maze systematically and find the exit without prior knowledge. A model-based agent with UCS is the most effective combination for this problem as it balances optimality, time and space usage, ensuring efficient and reliable decision-making.